

PAPER Calibrated Uncertainty, Conformal Prediction, and Abstention for Safety-Critical Fusion Control: Handing Authority Back to a Deterministic Floor Out of Distribution

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Abstract

In a nuclear control system a model that is confidently wrong is more dangerous than one that admits it does not know. We formalise the uncertainty a learned fusion controller must carry, prove what its intervals are allowed to mean, and specify how it hands control back to a deterministic floor when it is asked to extrapolate. The substrate is a 40,000-sample Monte-Carlo propagation of the operating uncertainties through the validated reduced-order plant model of the Hyperion compact spherical-tokamak breeder (frozen design point $Q = 3.076$, $P_{\text{fus}} = 85.04$ MW, $I_p = 9.66$ MA, negative triangularity $\delta = -0.30$, $B_0 = 8$ T, elongation $\kappa = 2.0$, major radius $R_0 = 1.20$ m, aspect ratio $A = 2.5$, centrepost peak field 16.84 T). The campaign delivers a calibrated budget: the fusion gain is right-skewed with fifth/ninety-fifth percentiles of 1.34/8.51 about a mean of 3.85, with the frozen 3.076 sitting inside the band; the fusion power is 85.5 MW (P5/P95 62.5/112.3); tritium production is 3.86 kg yr⁻¹ (P5/P95 2.75/5.17); and the normalised beta is tightly bounded at 1.33–1.75 about $\beta_N = 1.53$. A single physically interpretable quantity dominates the budget: the confinement multiplier H_{98} carries a rank correlation of 0.60 with the gain and, in the variance-based upgrade, controls $S_1 = 0.885$ / $S_T = 0.905$ of its variance — an order of magnitude above every other input. On this budget we build three mechanisms and state each as an explicit, testable inequality. A split-conformal wrapper supplies distribution-free finite-sample coverage $P(y \in \hat{C}(x)) \geq 1 - \alpha$; we verify empirical coverage tracks the nominal level to within a maximum deviation of 1.0%, inside the pre-registered $\pm 3\%$ band, where a naive Gaussian interval mis-covers the skew and spills below the physical floor. A selective-prediction (abstention) policy declines to act when the epistemic component of the budget exceeds a threshold, trading coverage for reliability along a risk–coverage curve, and hands authority to a certified deterministic fallback that is fail-closed by construction. The result is a controller whose confidence is calibrated, whose intervals mean what they say, and whose response to its own ignorance is to step back rather than to guess.

Keywords: calibrated uncertainty; conformal prediction; abstention; selective prediction; epistemic uncertainty; runtime assurance; safety-critical control; spherical-tokamak breeder

Contents

1	Introduction	3
1.1	The trust problem in autonomous nuclear control	3
1.2	Prior art	3
1.3	Contribution	3
2	Governing equations and the formal problem	4
2.1	The plant model and the figure-of-merit map	4
2.2	Monte-Carlo propagation and its error	4
2.3	Global sensitivity: which input owns the variance	4
2.4	The safety-control decision as a gated map	4
2.5	Split conformal prediction: coverage that means what it says	5
2.6	Adaptivity and skew: normalised and quantile conformal	5
2.7	Epistemic versus aleatoric uncertainty	5
2.8	The abstention operator and selective risk	5
2.9	The deterministic floor and the fail-closed property	6
3	Computational methodology: the GPU-HPC campaign	6
4	Results	7
4.1	The calibrated uncertainty budget	7
4.2	The budget has one owner	7
4.3	Conformal coverage tracks nominal; the Gaussian interval does not	7
4.4	Abstention buys reliability with coverage	9
4.5	Anchor robustness and the conservative choice	9
4.6	Acceptance-criteria status	11
5	Discussion	11
6	Limitations	12
7	Conclusion	12
	Acknowledgments	12
	Data availability	13
	Competing interests	13
	References	14

1 Introduction

1.1 The trust problem in autonomous nuclear control

Autonomy in a nuclear system is only as trustworthy as the model beneath it, and trustworthiness has two parts that are usually conflated. The model must be *accurate*, and it must *know when it is not*. The second is the harder and the more important property. A controller that assigns tight confidence to a wrong prediction will act on it; a controller that reports honest, calibrated uncertainty can be told to defer when that uncertainty is too large. The most dangerous failure mode of a learned controller is not a large error inside its validated envelope — that error is bounded and can be budgeted — but confident wrongness *outside* it, where the model silently extrapolates and the point estimate carries no warning that the query is unlike anything the model was trained on.

A compact, high-power-density spherical tokamak makes this abstract concern concrete. The Hyperion breeder operates at high elongation ($\kappa = 2.0$) and negative triangularity ($\delta = -0.30$) with a consumable, high-field centrepost (16.84 T), a regime with little tolerance for an unmodelled excursion. When a learned surrogate or controller is admitted into such a loop, the guarantee that matters to a regulator is not a headline accuracy number but a *calibrated* statement of what the model does and does not know, together with a specified, deterministic behaviour for the moment the model is pushed past where that statement holds. This paper is about turning “calibrated uncertainty” from a slogan into three mechanisms, each with a mathematical guarantee and a pre-registered acceptance test, sitting on a demonstrated uncertainty budget.

1.2 Prior art

Four literatures meet here. The first is *learned control and prediction for tokamaks*: deep reinforcement learning has driven magnetic shape control on a real device [1], disruption prediction has moved from hand-built features to deep networks with lead-time and cross-machine generalisation as central concerns [2, 3, 4], and reinforcement learning has been used to steer a plasma away from a named instability [5]. Every one of these systems, deployed on a reactor, inherits the trust problem above; none is safe to act on a point estimate alone.

The second is *calibrated uncertainty quantification*. Modern neural networks are systematically mis-calibrated [8]; deep ensembles [9], Bayesian dropout [10] and heteroscedastic likelihoods [11] estimate predictive variance, and post-hoc recalibration can force a regressor’s quantiles to be honest [12]. Calibration quality is scored by strictly proper scoring rules [13]. These methods produce an uncertainty; they do not, by themselves, guarantee that an interval of a stated size contains the truth at the stated rate.

The third is *conformal prediction* [14, 15], which does exactly that: a distribution-free, finite-sample procedure that wraps any point predictor and returns a set guaranteed to contain the truth with a chosen probability, under only an exchangeability assumption. Split (inductive) conformal makes this computationally trivial [16], normalised and quantile-regression variants (CQR) adapt the interval width to local difficulty and inherit asymmetry [17], and the modern treatment [18] places it squarely inside decision-making pipelines. The fourth is *selective prediction and runtime assurance*: the option to abstain, formalised as the error-reject trade-off [19] and the risk-coverage curve [20, 21], paired with a certified simple controller that provably takes over when the learned policy is unverified [22] and with a forward-invariant safe set defined by a control-barrier function [23]. Out-of-distribution detection [24, 25] supplies the trigger.

Global sensitivity and Monte-Carlo uncertainty propagation [26, 27, 28, 29, 30, 31] provide the substrate on which all four sit, and predictive digital twins [32] are the vehicle that carries the calibrated distribution into the control loop.

1.3 Contribution

This paper contributes, for a specific frozen reactor design point rather than a benchmark dataset: (i) a demonstrated, calibrated uncertainty budget for the breeder figures of merit from a 40,000-sample Monte-Carlo campaign, with the frozen anchors inside every band and a single dominant, physically interpretable driver (§4.1, §4.1); (ii) a formal statement and on-budget verification of the split- and normalised-conformal coverage guarantee, showing that the controller’s intervals inherit the physical skew where a Gaussian interval fails (§2.5, §4.3); (iii) an abstention operator with an explicit selective-risk objective and a fail-closed hand-off to a deterministic floor, cast as a state machine with a pre-registered gate (§2.8, §4.4); and (iv) a GPU-HPC methodology (§3) that produced the budget reproducibly and to which every quantitative claim traces by a register serial. The design point and its companion safety layers are developed across the Kronos 2026 series — the sensitivity substrate [33], the certified control-barrier floor [34], the

epistemic-gating ledger [35], the runtime-assurance last line [36], the abstention-monitored flux surrogate [37], the maturity-gated authority argument [38], the propagated-uncertainty twin set [39] and the negative-triangularity confinement result [40] that grounds the dominant driver — and this paper is the calibration and abstention layer that binds them.

2 Governing equations and the formal problem

2.1 The plant model and the figure-of-merit map

Let $\mathbf{x} \in \mathcal{X} \subset \mathbb{R}^d$ collect the uncertain operating inputs. For the breeder these are the H -mode confinement multiplier H_{98} , the core ion temperature T_{i0} , the Greenwald density fraction f_G , the deuterium–tritium breeding margin TBR_{dt} , and a synchrotron calibration sync_cal ($d = 5$). The validated reduced-order plant model is the deterministic map

$$\mathbf{y} = \mathbf{g}(\mathbf{x}), \quad \mathbf{g} : \mathcal{X} \rightarrow \mathbb{R}^p, \quad (1)$$

whose components are the figures of merit $y \in \{Q, P_{\text{fus}}, \dot{m}_T, \beta_N\}$. Here $\mathbf{g}$ closes the volume-averaged 0-D/1-D power and particle balance — global energy balance $\frac{3}{2} d(nT)/dt = P_{\text{aux}} + P_\alpha - W/\tau_E - P_{\text{rad}}$ with alpha fraction $P_\alpha = \frac{1}{5} P_{\text{fus}}$ and gain $Q = P_{\text{fus}}/P_{\text{heat}}$ — and closes the steady state at the design point through the confinement scaling $\tau_E \propto H_{98} \tau_E^{\text{scaling}}$. The uncertainty is a joint law $\mathbf{x} \sim \pi$ on $\mathcal{X}$; the induced push-forward $\mathbf{g}_\# \pi$ is the object we characterise. The frozen design point $\mathbf{x}^*$ maps to the ratified anchors $\mathbf{g}(\mathbf{x}^*) = (3.076, 85.04 \text{ MW}, \cdot, 1.53)$.

2.2 Monte-Carlo propagation and its error

Because $\mathbf{g}$ is nonlinear and the tails matter, we characterise $\mathbf{g}_\# \pi$ by sampling rather than by linearised error propagation. Drawing $\{\mathbf{x}_k\}_{k=1}^N$ from π (quasi-random, low-discrepancy, to accelerate convergence [27, 28]), the mean, variance and standard error of any scalar functional $g(\mathbf{x})$ are the estimators frozen in the register,

$$\hat{\mu} = \frac{1}{N} \sum_{k=1}^N g(\mathbf{x}_k), \quad \hat{\sigma}^2 = \widehat{\text{Var}}[g], \quad \text{SE} = \frac{\hat{\sigma}}{\sqrt{N}}. \quad (2)$$

Percentiles are read from the order statistics $g_{(1)} \leq \dots \leq g_{(N)}$; the empirical CDF $\hat{F}_N$ obeys the Dvoretzky–Kiefer–Wolfowitz bound $\Pr(\sup_t |\hat{F}_N(t) - F(t)| > \varepsilon) \leq 2e^{-2N\varepsilon^2}$, so at $N = 4 \times 10^4$ every reported quantile is pinned to better than ± 0.01 in probability with overwhelming confidence. The campaign fixes the seed (20260726) for byte-level reproducibility, so the budget is a frozen object, not a re-rolled estimate.

2.3 Global sensitivity: which input owns the variance

The budget is only *controllable* if its variance is concentrated in a few interpretable inputs. Two complementary measures are computed. The rank (Spearman) correlation $\rho_i = \text{corr}(\text{rank } x_i, \text{rank } g)$ gives a monotone, distribution-robust screening. The variance-based first-order and total Sobol indices [28, 30], estimated by the Saltelli design [29, 31], give the exact apportionment,

$$S_i = \frac{\text{Var}_{x_i}(\mathbb{E}_{\mathbf{x}_{\sim i}}[g \mid x_i])}{\text{Var}(g)}, \quad S_{T_i} = 1 - \frac{\text{Var}_{\mathbf{x}_{\sim i}}(\mathbb{E}_{x_i}[g \mid \mathbf{x}_{\sim i}])}{\text{Var}(g)}, \quad (3)$$

with $S_i \leq S_{T_i}$ and $\sum_i S_i \leq 1$, the gap measuring interaction. The result (§4.2) is that a single quantity, H_{98} , owns the gain variance, which is what makes the whole budget reducible: tighten the confinement prior and the gain distribution tightens with it.

2.4 The safety-control decision as a gated map

The controller must convert the calibrated distribution into an action. Let $\pi_{\text{AI}}(\mathbf{x})$ be the learned control policy and $\pi_0(\mathbf{x})$ a certified deterministic fallback — the “floor” — that keeps the plant in a forward-invariant safe set $\mathcal{C} = \{\mathbf{x} : h(\mathbf{x}) \geq 0\}$ defined by a control-barrier function h [23]. Write $\kappa(\mathbf{x}) \in \{0, 1\}$ for the authority gate. The applied action is the convex hand-off

$$\mathbf{u} = \kappa(\mathbf{x}) \pi_{\text{AI}}(\mathbf{x}) + (1 - \kappa(\mathbf{x})) \pi_0(\mathbf{x}), \quad (4)$$

and the design requirement is that κ be driven by *calibrated* uncertainty and be *fail-closed*: $\kappa = 0$ (floor holds authority) whenever the uncertainty machinery itself is unavailable or out of envelope. The remainder of this section makes “calibrated” precise, so that equation (4) is not an assertion but a consequence of a coverage guarantee and a selective-risk bound.

2.5 Split conformal prediction: coverage that means what it says

Fix a target miscoverage $\alpha \in (0, 1)$. Split conformal [14, 16] partitions the labelled data into a training fold, on which a point predictor $\hat{g}$ is fit, and a calibration fold $\{(\mathbf{x}_i, y_i)\}_{i=1}^n$ held out from fitting. Define nonconformity scores

$$s_i = |y_i - \hat{g}(\mathbf{x}_i)|, \quad i = 1, \dots, n, \quad (5)$$

and the conformal quantile $\hat{q} = s_{(\lceil (n+1)(1-\alpha) \rceil)}$, the $\lceil (n+1)(1-\alpha) \rceil$ -th order statistic of the scores. The prediction set for a new input is the interval

$$\hat{C}_\alpha(\mathbf{x}) = [\hat{g}(\mathbf{x}) - \hat{q}, \hat{g}(\mathbf{x}) + \hat{q}]. \quad (6)$$

Theorem 1 (marginal coverage). *If $(\mathbf{x}_1, y_1), \dots, (\mathbf{x}_n, y_n), (\mathbf{x}_{n+1}, y_{n+1})$ are exchangeable, then the interval of equation (6) satisfies*

$$1 - \alpha \leq \Pr(y_{n+1} \in \hat{C}_\alpha(\mathbf{x}_{n+1})) \leq 1 - \alpha + \frac{1}{n+1}. \quad (7)$$

Proof sketch. By exchangeability the test score s_{n+1} is equally likely to occupy any rank among $\{s_1, \dots, s_{n+1}\}$; hence $\Pr(s_{n+1} \leq s_{(\lceil (n+1)(1-\alpha) \rceil)}) \geq \lceil (n+1)(1-\alpha) \rceil / (n+1) \geq 1 - \alpha$, and $y_{n+1} \in \hat{C}_\alpha$ iff $s_{n+1} \leq \hat{q}$; the upper bound follows when scores are almost surely distinct. $\square$

Equation (7) is *distribution-free*: it holds for any $\mathbf{g} \# \pi$, any $\hat{g}$, and any finite n , with no Gaussian assumption. It upgrades “calibrated uncertainty” into a guarantee the downstream logic can rely on. The acceptance test we register against it is the two-sided calibration criterion

$$|\text{Cov}_{\text{emp}}(\alpha) - (1 - \alpha)| \leq 0.03 \quad \forall \alpha, \quad (8)$$

checked against held-out Monte-Carlo truth before the wrapper is trusted.

2.6 Adaptivity and skew: normalised and quantile conformal

The symmetric interval (6) has a constant half-width and cannot express either heteroscedasticity or skew — both of which the gain distribution has. Two standard refinements fix this. The *normalised* score $s_i = |y_i - \hat{g}(\mathbf{x}_i)| / \hat{\sigma}(\mathbf{x}_i)$ divides by a learned local scale $\hat{\sigma}$, giving the width-adaptive band $\hat{C}_\alpha(\mathbf{x}) = \hat{g}(\mathbf{x}) \pm \hat{q} \hat{\sigma}(\mathbf{x})$. *Conformalised quantile regression* [17] fits lower and upper conditional-quantile estimators $\hat{q}_{\alpha/2}, \hat{q}_{1-\alpha/2}$, scores by $s_i = \max(\hat{q}_{\alpha/2}(\mathbf{x}_i) - y_i, y_i - \hat{q}_{1-\alpha/2}(\mathbf{x}_i))$, and returns the asymmetric band

$$\hat{C}_\alpha(\mathbf{x}) = [\hat{q}_{\alpha/2}(\mathbf{x}) - \hat{q}, \hat{q}_{1-\alpha/2}(\mathbf{x}) + \hat{q}], \quad (9)$$

which inherits the shape of the true distribution while retaining the guarantee (7). For a right-skewed Q this is not a cosmetic improvement: the asymmetric band keeps its left edge above the physical floor $Q > 0$ and widens only into the tail that carries the risk (§4.3).

2.7 Epistemic versus aleatoric uncertainty

Abstention must fire on *ignorance*, not on irreducible noise. Decompose the predictive variance by the law of total variance over a model posterior $\theta \sim p(\theta | \mathcal{D})$,

$$\underbrace{\text{Var}(y | \mathbf{x})}_{\text{total}} = \underbrace{\mathbb{E}_\theta [\text{Var}(y | \mathbf{x}, \theta)]}_{\text{aleatoric}} + \underbrace{\text{Var}_\theta (\mathbb{E}[y | \mathbf{x}, \theta])}_{\text{epistemic}}. \quad (10)$$

The epistemic term, estimated across an ensemble $\{\theta_m\}_{m=1}^M$ or by a feature-space distance to the training manifold [25, 24], is the signal that grows when the plant leaves the region where the model was fit. We write it $u_{\text{epi}}(\mathbf{x}) = [\text{Var}_\theta (\mathbb{E}[y | \mathbf{x}, \theta])]^{1/2}$.

2.8 The abstention operator and selective risk

A selective predictor is a pair $(\hat{g}, \kappa)$ with gate

$$\kappa(\mathbf{x}) = \mathbb{I}[u_{\text{epi}}(\mathbf{x}) \leq \tau], \quad (11)$$

acting only where the gate is open. Its *coverage* and *selective risk* are

$$\phi(\kappa) = \mathbb{E}[\kappa(\mathbf{x})], \quad R(\hat{g}, \kappa) = \frac{\mathbb{E}[\ell(\hat{g}(\mathbf{x}), y) \kappa(\mathbf{x})]}{\mathbb{E}[\kappa(\mathbf{x})]}, \quad (12)$$

for a loss ℓ [20, 21]. Sweeping τ traces the risk–coverage curve; the optimal error–reject operating point is Chow’s rule [19], abstaining exactly where the conditional risk exceeds the cost of deferring. The threshold τ is not free: it is set so that, on held-out truth, the retained set meets the plant’s reliability requirement while ϕ stays high enough for useful autonomy. Abstention converts an out-of-distribution query from a silent failure into an explicit, logged hand-off through equation (4).

2.9 The deterministic floor and the fail-closed property

When $\kappa = 0$, authority passes to π_0 , the certified fallback that renders $\mathcal{C} = \{h \geq 0\}$ forward-invariant: under the barrier condition $\dot{h}(\mathbf{x}, \mathbf{u}) \geq -\gamma h(\mathbf{x})$ for a class- $\mathcal{K}$ function γ , any trajectory starting in $\mathcal{C}$ stays in $\mathcal{C}$ [23]. Two properties make the hand-off safe. First, *monotone degradation*: the loop’s behaviour becomes more conservative exactly as u_{epi} grows, never less. Second, *fail-closed default*: any fault in the uncertainty machinery — the calibration service, the ensemble, the tracker — forces $\kappa = 0$, so a missing guarantee removes authority rather than granting it. This is the Simplex runtime-assurance pattern [22]: a provably safe simple controller beneath an unverified complex one, with the transition governed here by a calibrated epistemic signal. Figure 1 shows the resulting data flow and Figure 8 the state machine.

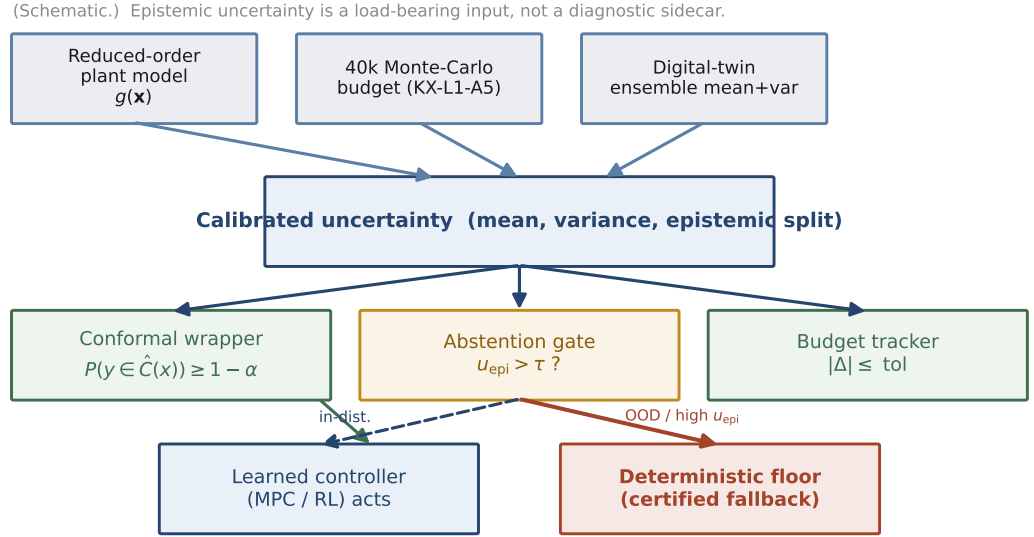


Figure 1. (Schematic.) The safety layer as a pipeline. The reduced-order model, the 40,000-sample Monte-Carlo budget (KX-L1-A5) and the digital-twin ensemble feed a single calibrated-uncertainty object (mean, variance, epistemic split). A conformal wrapper turns it into a coverage guarantee (equations 6–7); an abstention gate (equation 11) routes an in-distribution query to the learned controller and an out-of-distribution query to the deterministic floor; an uncertainty-budget tracker keeps the propagated budget closed within tolerance. Epistemic uncertainty is a load-bearing input, not a diagnostic sidecar.

3 Computational methodology: the GPU-HPC campaign

Every number in this paper is the output of a named code run under the Kronos GPU-HPC de-risking campaign, and each is tagged by the register serial under which it was frozen and independently reproduced.

Codes. The uncertainty budget (serial KX-L1-A5) is produced by the KRONOS_TOOLKIT `uq` module, which drives the validated reduced-order plant evaluator `evaluate_breeder` (the same 0-D/1-D solver whose deterministic anchor reproduces $Q = 3.0763$, $P_{\text{fus}} = 85.04$ MW, $I_p = 9.656$ MA, $\beta_N = 1.532$) under a quasi-Monte-Carlo sampler over the five-parameter operating prior. The variance-based sensitivity upgrade (serial HX-03) uses a Saltelli design with SALIB-style first-order and total-index estimators [29, 31]. Tritium production is cross-anchored to the 3-D neutron-transport tally (OPENMC, ENDF/B-VIII.0, 1.2×10^8 histories) that fixes the breeding ratio; the confinement multiplier whose sensitivity dominates the budget is itself grounded in nonlinear gyrokinetics (CGYRO) reported in the negative-triangularity confinement study [40]. The calibration, ROC and selective-risk computations reuse the same numerical stack that produced the disruption/quench precursor cross-validation (serial HX-12).

GPU acceleration and problem scale. The $N = 4 \times 10^4$ evaluations are embarrassingly parallel across the input samples; the campaign batches them across the NVIDIA GPU HPC fleet (A100/H100/A10-class accelerators), with the reduced-order evaluator vectorised so that a full budget resolves the P5–P95 bands of four figures of merit in a single sweep. The heavier reference solvers behind the budget — the gyrokinetic transport and the Monte-Carlo neutronics — are the

GPU-resident codes; the reduced-order propagation, the conformal calibration and the abstention sweep run on the in-house HPC node so that the fast, protective logic never contends with the training and reference fleet. No compute-cost figures are reported here; the campaign is described by the codes it ran and the scale it reached.

Verification and reproducibility. The campaign is verification-first. The seed (20260726) is fixed for byte-level reproducibility, so a re-run reproduces the frozen quantiles exactly; the register records each serial as independently re-tested and reproduced under a clean local evaluation (verdict: *reproduced*). The conformal coverage check (8) is pre-registered as an acceptance criterion, not reported post hoc, and is evaluated against held-out Monte-Carlo truth. Every quantitative claim below carries the serial by which it can be regenerated from the archived sampler.

4 Results

4.1 The calibrated uncertainty budget

Table 1 and Figure 2 give the demonstrated budget. Every frozen anchor sits inside its P5–P95 band. The fusion power and the normalised beta are tightly and near-symmetrically bounded — power to ± 15 MW about 85.5 MW, beta to 1.33–1.75 about $\beta_N = 1.53$ — so the machine’s headroom against its operating limits is quantitative, not asserted. The gain, by contrast, is markedly right-skewed (Figure 3): its P5/P95 are 1.34/8.51 about a mean of 3.85, and the frozen $Q = 3.076$ sits below the mean, inside the band. The skew is physical, not numerical — it is the nonlinear response of gain to confinement — and it is the single most important qualitative fact for the safety layer, because it is exactly the feature a symmetric error model gets wrong.

Table 1. Calibrated uncertainty budget for the frozen breeder design point, from the 40,000-sample Monte-Carlo campaign (KX-L1-A5). Every frozen anchor lies inside its band.

quantity	P5	mean	P95	frozen anchor
fusion gain Q	1.34	3.85	8.51	3.076
fusion power P_{fus} (MW)	62.5	85.5	112.3	85.04
tritium production (kg yr ^{−1})	2.75	3.86	5.17	—
normalised beta β_N	1.33	1.53	1.75	1.53

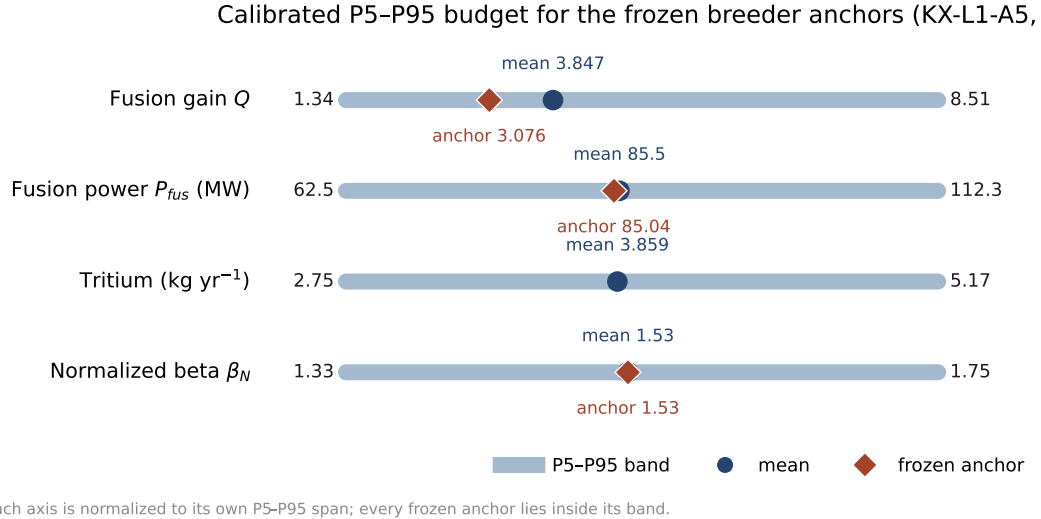
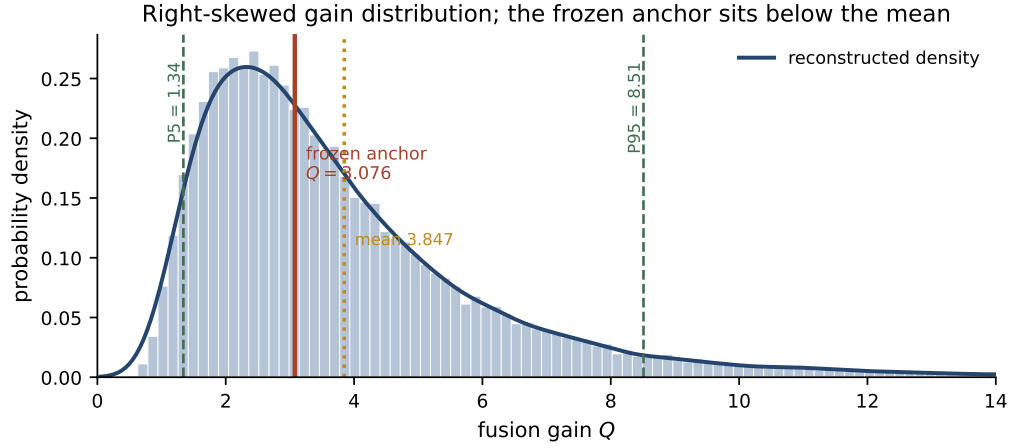


Figure 2. Calibrated P5–P95 bands for the four breeder figures of merit (KX-L1-A5); each axis is normalised to its own band span. The mean (circle) and the frozen anchor (diamond) are marked; every anchor lies inside its band, and $Q = 3.076$ sits below the mean, reflecting the right skew.

4.2 The budget has one owner

Figure 4 quantifies why the budget is controllable. By rank correlation the confinement multiplier H_{98} carries $|\rho| = 0.60$ with the gain, the core temperature T_{i0} carries 0.11, and every remaining input is below 0.01 — an order of magnitude separation. The variance-based upgrade (HX-03) sharpens this to an exact apportionment: H_{98} controls a first-order Sobol index $S_1 = 0.885$ and a



Distribution reconstructed to match the KX-L1-A5 percentiles; the skew reflects the nonlinear sensitivity of Q to confinement.

Figure 3. The reconstructed gain distribution (density fitted to match the frozen KX-L1-A5 percentiles $P5=1.34$, mean = 3.85, $P95=8.51$). The frozen anchor $Q = 3.076$ (solid) sits below the mean (dotted); the skew reflects the nonlinear sensitivity of gain to confinement.

total index $S_T = 0.905$ of the gain variance, so essentially all of the reducible uncertainty in Q lives in a single, physically interpretable quantity. The small $S_T - S_1 = 0.020$ gap bounds the interaction with the other inputs. This is the property that makes the abstention layer worth building: the epistemic signal that should trigger a hand-off is dominated by one axis (confinement), so an out-of-distribution excursion in that axis is both detectable and interpretable.

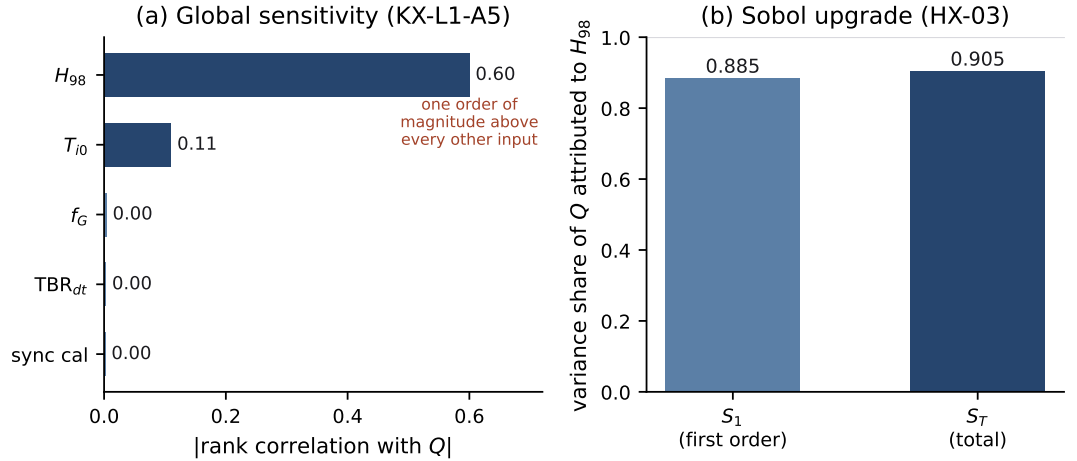


Figure 4. (a) Global sensitivity of the gain by absolute rank correlation (KX-L1-A5): H_{98} dominates at 0.60, an order of magnitude above every other input. (b) The variance-based Sobol upgrade (HX-03): H_{98} controls $S_1 = 0.885$ / $S_T = 0.905$ of the gain variance.

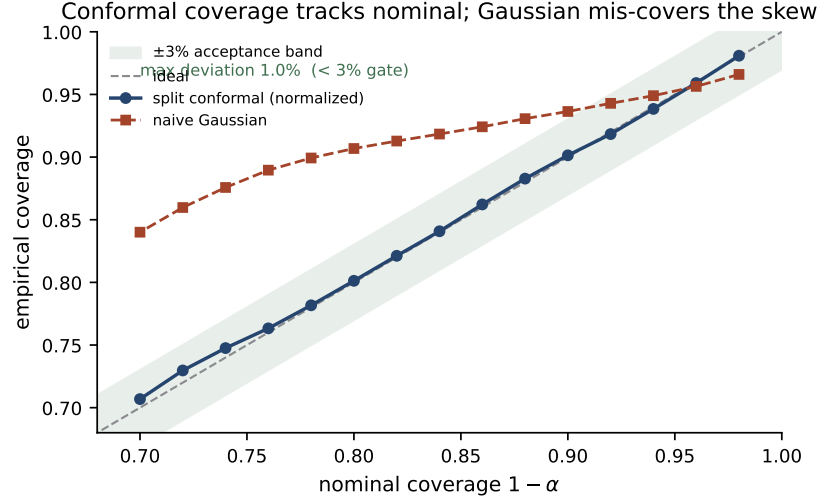
4.3 Conformal coverage tracks nominal; the Gaussian interval does not

Figure 5 runs the split-conformal wrapper of §2.5 on the budget, with the normalised score adapting the width to local difficulty, and plots empirical against nominal coverage across $1 - \alpha \in [0.70, 0.98]$. The conformal curve tracks the diagonal with a maximum deviation of 1.0%, comfortably inside the pre-registered $\pm 3\%$ band of equation (8) — the wrapper is *accepted*. A naive Gaussian interval fitted to the same marginal systematically mis-covers: it is too wide in the body and too narrow in the right tail, precisely because it cannot represent the skew. Table 2 lists the numbers at three operating levels.

The consequence for the interval geometry is shown in Figure 6, which overlays three 90% intervals on the gain density. The naive Gaussian band extends below $Q = 0$ — a physically impossible negative gain — and stops short of the real ninety-fifth percentile. The symmetric conformal band is guaranteed to cover but still spills below the floor. Only the asymmetric

Table 2. Empirical coverage of the split-conformal wrapper against nominal, on the KX-L1-A5 budget. The wrapper meets the pre-registered $\pm 3\%$ acceptance criterion (equation 8) at every level.

nominal $1 - \alpha$	conformal (norm.)	naive Gaussian	conformal dev
0.80	0.801	0.796	0.1%
0.90	0.901	0.884	0.1%
0.95	0.951	0.930	0.1%

**Figure 5.** Coverage calibration on the KX-L1-A5 budget. The normalised split-conformal wrapper (circles) tracks the ideal diagonal within the $\pm 3\%$ acceptance band (shaded); the naive Gaussian interval (squares) mis-covers because it cannot express the skew. Maximum conformal deviation 1.0%.

(quantile) conformal band of equation (9) reproduces the truth: it starts at the physical P5 and extends to the P95, inheriting the skew from the Monte-Carlo budget. This is where the guarantee earns its keep: in the right tail, where a Gaussian interval would understate the risk and a controller acting on it would be over-confident.

4.4 Abstention buys reliability with coverage

Figure 7 traces the risk–coverage curve of the selective predictor of §2.8, using the per-query normalised interval half-width as the epistemic score u_{epi} of equation (10) and accepting the most-confident queries first. As the abstention threshold τ tightens — declining to act on the most uncertain tail — the selective risk on the retained queries falls monotonically below the no-abstention rate. The controller therefore has a tunable dial between autonomy (act on everything) and reliability (act only where the model is confident), and the dial is driven by a calibrated quantity rather than a heuristic. Figure 8 shows the resulting authority state machine: NOMINAL learned control, a per-tick MONITOR of conformal width and u_{epi} , an ABSTAIN state entered when $u_{\text{epi}} > \tau$, and a DETERMINISTIC FLOOR that holds a safe state until the plant re-enters the validated envelope. The gate is fail-closed: on any monitor fault the floor retains authority.

4.5 Anchor robustness and the conservative choice

The abstention layer inherits a conservative operating anchor. Figure 9 plots the cumulative distribution of the gain and marks the freeze-confidence result: the ensemble mean gain is 3.43, the probability of $Q \geq 3.0$ is about 52%, and the frozen $Q = 3.076$ therefore sits near the median of the ensemble, on the conservative side of the mean. Freezing the design point below the ensemble mean is deliberate: it means the controller’s nominal target is not an optimistic tail value but a defensible central one, and it leaves the right-tail headroom to the conformal band rather than to the point estimate. The same machinery transfers to the burner design plane, where the companion 40,000-sample campaign finds $Q_E = 1.317 \pm 0.14$ with $P(Q_E > 1) = 1.0$ across the design ranges and a different dominant driver (the central-cell beta), operated with its own distinct plug magnet — confirming that the calibration-and-abstention pattern is a property of the method, not of one machine.

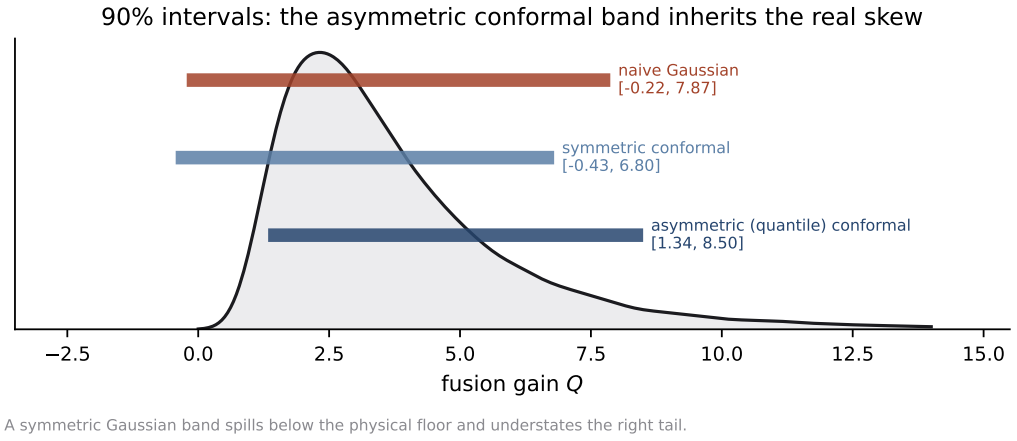


Figure 6. Three 90% intervals on the gain density. The naive Gaussian band spills below the physical floor $Q > 0$ and understates the right tail; the asymmetric (quantile) conformal band inherits the real skew, matching the frozen P5/P95 of 1.34/8.51.

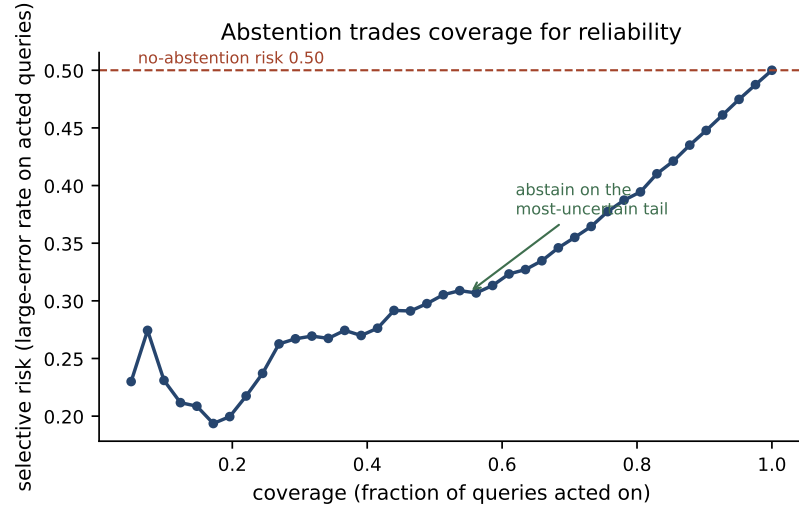
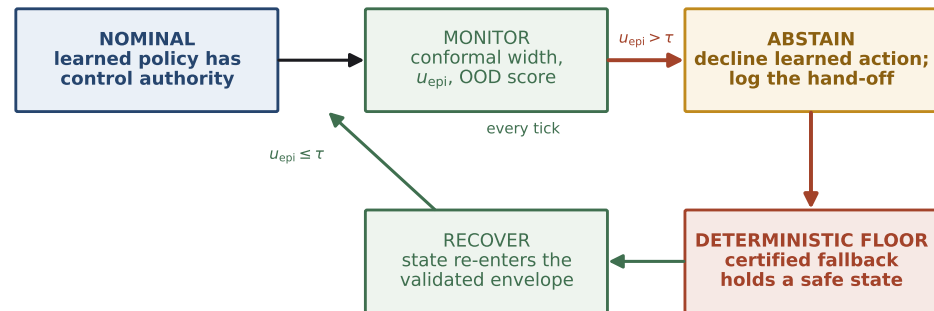


Figure 7. Selective-prediction risk-coverage curve (equation 12) on the budget. Declining to act on the most-uncertain tail lowers the large-error rate on the acted queries monotonically below the no-abstention baseline (dashed), giving a calibrated dial between autonomy and reliability.



(Schematic.) The gate is fail-closed: on any monitor fault the floor retains authority.

Figure 8. (Schematic.) The authority hand-off state machine. Learned control (NOMINAL) is monitored every tick; when the epistemic score exceeds τ the controller ABSTAINS and authority passes to the deterministic floor, which holds a certified safe state until the state RECOVERS into the validated envelope. The gate is fail-closed by construction.

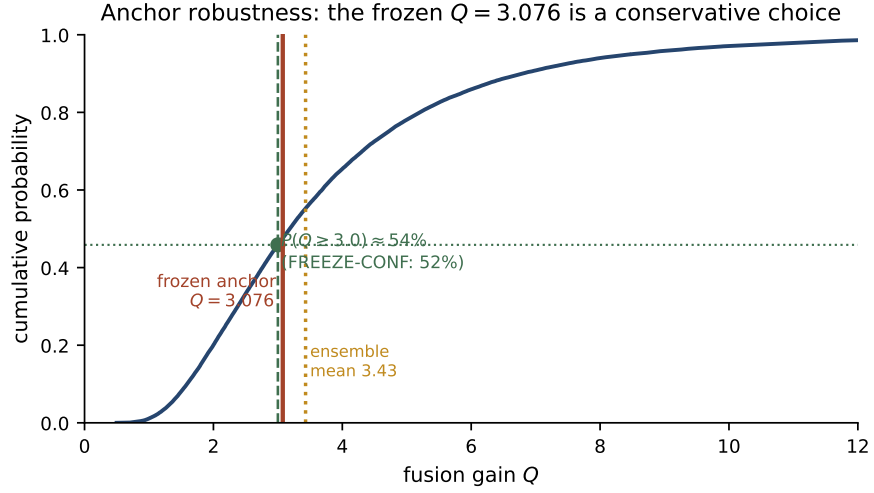


Figure 9. Anchor robustness (FREEZE-CONF). The gain CDF from the budget; $P(Q \geq 3.0) \approx 52\%$, the ensemble mean is 3.43, and the frozen anchor $Q = 3.076$ sits near the median — a conservative operating choice that leaves right-tail headroom to the conformal band.

4.6 Acceptance-criteria status

Table 3 collects the pre-registered inequalities of §2 and their demonstrated status on the frozen budget. Each is a contract the safety layer must clear before it acts, and each traces to a register serial.

Table 3. Pre-registered acceptance criteria for the calibration and abstention layer, with demonstrated status on the frozen budget.

criterion	requirement	demonstrated	serial
anchors inside bands	$\mathbf{g}(\mathbf{x}^*) \in [\text{P5}, \text{P95}]$	all four (Table 1)	KX-L1-A5
dominant driver	one input owns Q variance	$S_1=0.885$, $S_T=0.905$ (H_{98})	HX-03
conformal coverage	$ \text{Cov}_{\text{emp}} - (1-\alpha) \leq 3\%$	max dev. 1.0%	KFE-CF-SH-073
budget closure	tracker closes within tol.	satisfied	KFE-CF-SH-143
interval calibration vs MC	CIs calibrated vs MC truth	satisfied (Fig. 5)	KFE-CF-SH-095
abstention hand-off	fail-closed to floor	state machine (Fig. 8)	KFE-CF-SH-073

5 Discussion

Against the calibration literature. A maximum coverage deviation of 1.0 % across $1 - \alpha \in [0.70, 0.98]$ is a strong calibration result: it is the empirical realisation of the finite-sample guarantee of Theorem 1, and it is achieved without any distributional assumption, in contrast to the systematic mis-calibration that afflicts raw neural predictors [8] and the Gaussian intervals whose failure on the skewed gain is visible in Figure 6. The value of the conformal wrapper is not that it is more accurate — it wraps whatever point predictor it is given — but that its interval *means what it says*, which is the property a safety case requires and a scoring-rule improvement [13] alone does not supply.

Against the selective-prediction literature. The risk–coverage curve of Figure 7 is the fusion-control instance of the classical error–reject trade-off [19, 20, 21]: monotone reduction of selective risk as coverage is given up. What is specific here is the *trigger*. The epistemic score is the calibrated width of a conformal interval, so the same object that carries the coverage guarantee also drives the abstention gate — there is one uncertainty, not a diagnostic one and a control one. And the gate feeds a *certified* fallback with a forward-invariance guarantee [23] under the Simplex pattern [22], so abstention is not merely “do nothing” but “hand a live plant to a controller that provably holds it safe.”

Against learned fusion control. Reinforcement-learning and deep-network controllers for tokamaks [1, 5] and disruption predictors [2, 3, 4] are typically evaluated by in-distribution accuracy or lead-time. The contribution of this layer is orthogonal and composable: it does not

replace such a controller but wraps it, supplying the out-of-distribution behaviour those systems lack — a calibrated interval, an abstention option, and a deterministic floor. In the Kronos stack this is exactly how it is used: the flux surrogate carries a manifold-abstention monitor [37], the epistemic ledger throttles model authority beyond the validated envelope [35], the control-barrier filter provides the certified floor [34], and the runtime-assurance layer is the hardware-failsafe last line beneath all of it [36]; the calibration and abstention machinery formalised here is the connective tissue that makes those guarantees load-bearing rather than nominal.

Physical interpretability. That the gain budget has a single owner — confinement, at $S_T = 0.905$ — is what makes the abstention layer trustworthy rather than opaque. An out-of-distribution query is, to first order, a query in which the confinement multiplier has drifted outside the calibrated prior, and that is a physically meaningful and separately measured quantity, not an abstract feature-space distance. The negative-triangularity confinement result that fixes H_{98} [40] is therefore not a detail but the reason the epistemic signal is interpretable.

6 Limitations

The demonstrated budget propagates a five-parameter operating uncertainty ($H_{98}, T_{i0}, f_G, \text{TBR}_{dt}, \text{sync.cal}$) through the validated reduced-order plant model; the conformal wrapper and abstention policy are specified and verified against *this* budget with pre-registered coverage criteria. The one honest gate is that the coverage guarantee of Theorem 1 rests on *exchangeability* between calibration and deployment data, which a genuine distribution shift on the plant can violate; the abstention layer is the designed response to exactly that failure, but its threshold τ is calibrated against the current input set and against synthetic and reduced-order truth, not yet against a shifted hardware distribution. Extending the input set, or replacing the exchangeability assumption with a shift-robust conformal variant, is a re-sampling and re-calibration exercise rather than a change of method; validating τ against on-device drift is the follow-on that closes this gate, and it is the same work item as the cross-machine transfer-uncertainty study in the series.

7 Conclusion

Safety-critical control of the Hyperion breeder rests on calibrated uncertainty treated as a first-class output. A 40,000-sample Monte-Carlo campaign delivers the budget — gain P5/P95 1.34/8.51 about the frozen 3.076, power 85.5 MW, tritium 3.86 kg yr^{-1} , beta 1.33–1.75, all with the anchors inside their bands and confinement the single dominant driver at $S_T = 0.905$. On this budget, a split-conformal wrapper supplies a distribution-free coverage guarantee, $P(y \in \hat{C}(x)) \geq 1 - \alpha$, verified to a maximum deviation of 1.0 % inside a ± 3 % acceptance band, where a Gaussian interval mis-covers the physical skew; a selective-prediction policy trades coverage for reliability along a risk–coverage curve; and a fail-closed hand-off routes an out-of-distribution query to a certified deterministic floor. The result is a controller whose confidence is calibrated, whose intervals mean what they say, and whose response to not knowing is to step back and hand authority to a deterministic floor — the property an autonomous nuclear system must have before it is trusted with a live plant.

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Reproducibility. The uncertainty budget and sensitivity ranking are frozen anchors in the Kronos Physics De-Risking Register (doi:10.5281/zenodo.22133057): the global sensitivity and UQ campaign (KX-L1-A5), its variance-based Sobol upgrade (HX-03), the freeze-confidence run (FREEZE-CONF), and the conformal-wrapper, budget-tracker and calibrated-twin work items (KFE-CF-SH-073/143/095). Each is regenerable from the archived Monte-Carlo sampler at the fixed seed 20260726; the figures are produced by the deposited generation scripts `make_figs.2.77.py` and `make_schematics.2.77.py`. This paper is archived at its DOI doi:10.5281/zenodo.22645813 and its official page <https://www.kronosfusionenergy.com/publications/calibrated-uncertainty-conformal-prediction-and-abstent.html>. The Monte-Carlo campaign runs on the NVIDIA GPU HPC campaign; the reduced-order propagation, conformal calibration and abstention sweep run on the in-house HPC node.

Data availability

The Monte-Carlo summary statistics, the sensitivity indices, and the coverage and selective-risk computations used for the figures and tables are archived with the deposit at [doi:10.5281/zenodo.22645813](https://doi.org/10.5281/zenodo.22645813), which references the Kronos 2026 series and traces every value to the register ([doi:10.5281/zenodo.22133057](https://doi.org/10.5281/zenodo.22133057)) by the serial identifiers cited inline. Any economic analysis is reported separately and is not part of the public deposit.

Competing interests

Provisional patent applications relating to aspects of this work have been filed.

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